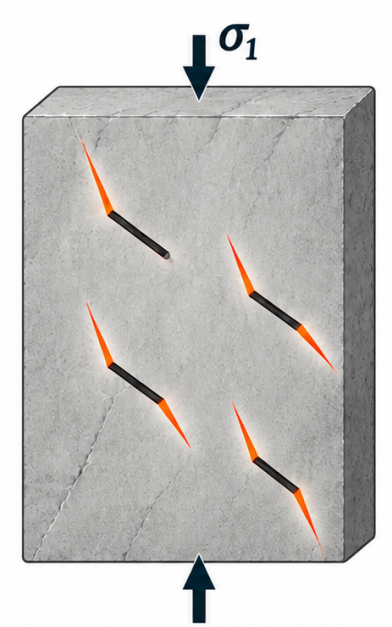
How do many small cracks interact?

An isolated-crack model is only the start of a many-crack failure process.

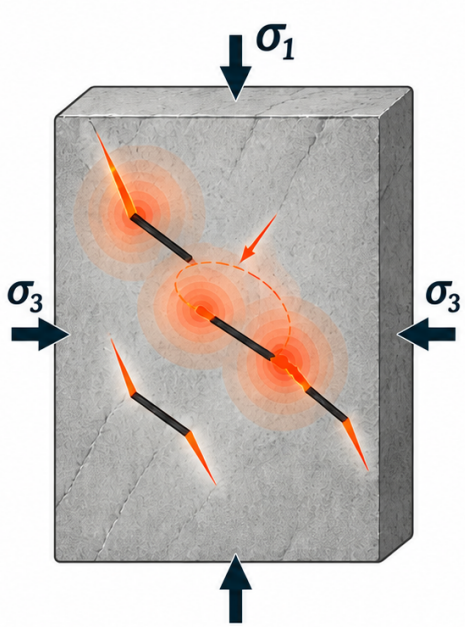
I · Isolated

Stable growth near favorably oriented flaws



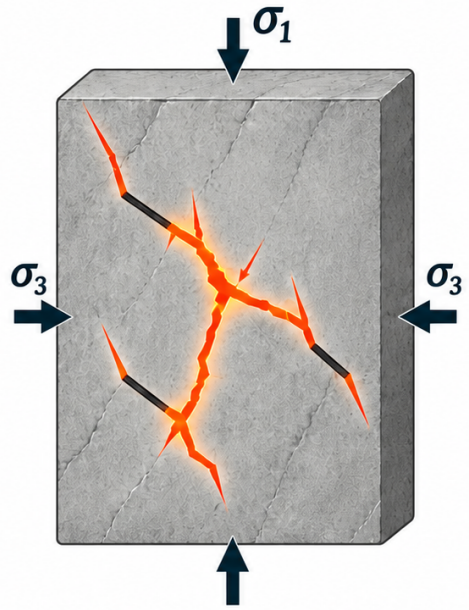
II · Interaction

Crack-tip stress fields overlap



III · Coalescence

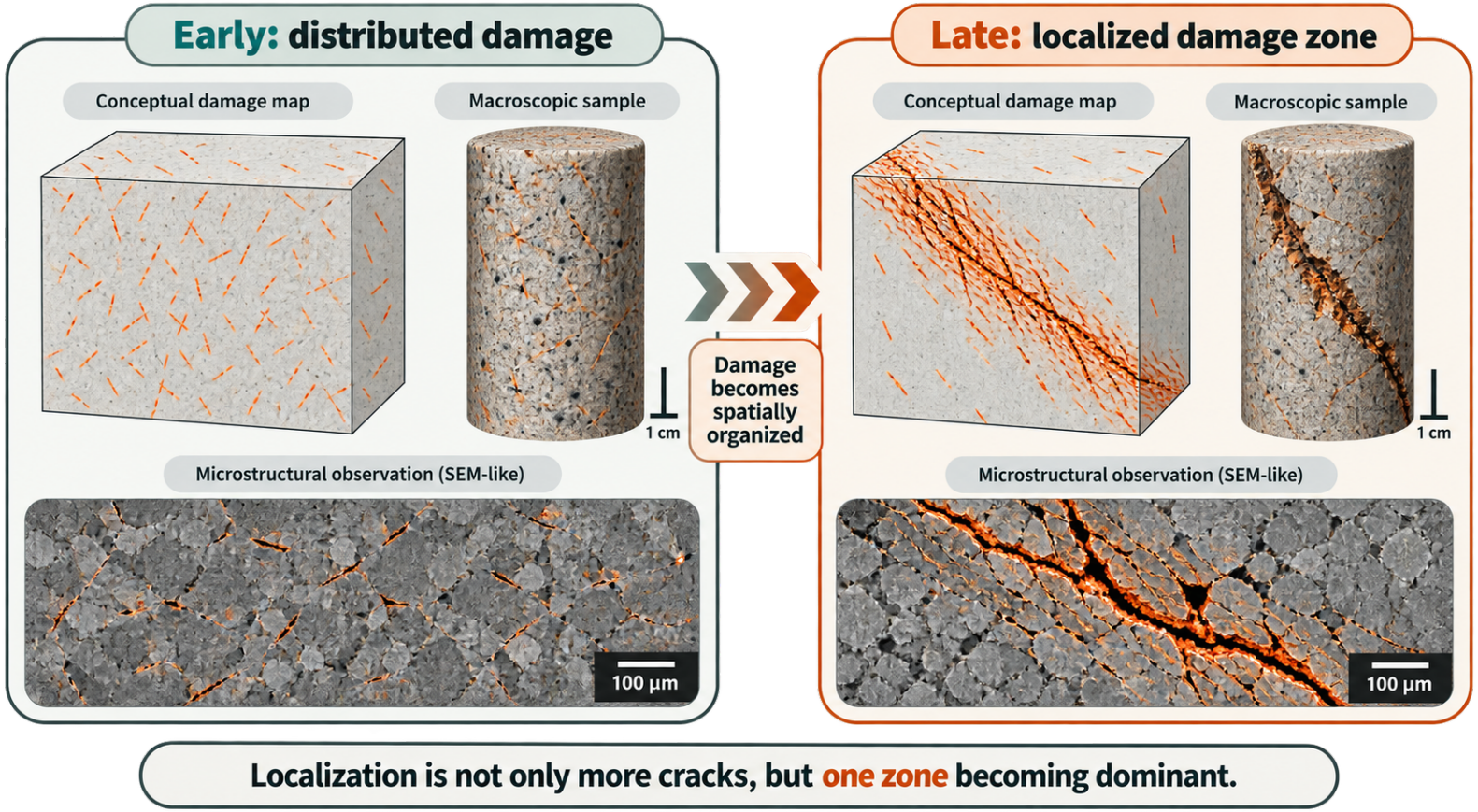
Connected fracture paths develop



Conceptual progression. Wong & Einstein (2009); Moore & Lockner (1995); McBeck et al. (2026).

Distributed microcracking becomes a localised fracture zone

The spatial organization of damage matters as much as the number of cracks.



Localization is not only more cracks, but one zone becoming dominant.

What would tell us that one zone is becoming dominant?

Conceptual damage maps; localization in experiments: Renard et al. (2018); McBeck et al. (2021, 2026).

From one flaw to a fault: the complete micromechanical story

1	2	3	4	5
Heterogeneity	**Sliding**	**Wing cracks**	**Interaction**	**Localization**
local stress concentration	friction threshold exceeded	local mode-I growth	coalescence and connectivity	system-spanning damage zone

Compact rock

The path is often dilatant and dominated by tensile microcracking around sliding flaws.

Porous rock

Pore collapse, grain crushing and compaction can join or replace the dilatant path.

Macroscopic shear failure emerges from many local events whose geometry and interaction evolve with load.

How can we detect and document the approach to failure?

A physical mechanism must be connected to an observable signal.

1 Initiation

Local tensile cracking

2 Growth

Distributed microcracking

3 Interaction

Cracks link and cluster

4 Localization

Macroscopic fault develops

What can we actually measure?

Stress–strain • **Active wave velocity** • **Acoustic emission signals**

Next: one microcracking event → an elastic wave → a sensor voltage → a detected AE event.

Conceptual synthesis across Paterson & Wong (Chs. 5–6) and laboratory AE studies.

How can we detect and document the approach to failure?

A physical mechanism must be connected to

1 Initiation

Local tensile cracking

Dis



4 Localization

Macroscopic fault develops

What can we actually m

Stress–strain • Active w **als**

Next: one microcracking event → an elas